

Carson Rivera

786-774-3620 | carsonrivera04@gmail.com | [linkedin.com/in/carsonrivera04](https://www.linkedin.com/in/carsonrivera04) | github.com/carsonrivera04 | Miami, FL

EDUCATION

Florida State University

Tallahassee, FL

Bachelor of Science in Computer Science - Honors Program | GPA: 3.88/4.00

May 2027

EXPERIENCE

Software Engineer Intern

June 2026 – Present

Merck & Co.

Rahway, NJ

- Developed a Python and FastAPI backend service for a RAG agent application, automating error routing to application owners to ensure uninterrupted delivery of critical SOP insights to R&D
- Resolved frontend React tickets and implemented automated Vitest component tests to improve UI stability and ensure consistent application performance during deployments.
- Partnered with application owners across R&D IT to drive the adoption of cost-effective cloud efficiency and FinOps practices for hosted applications.

Research Assistant - AI/ML

Sept. 2024 – Apr. 2026

Florida State University

Tallahassee, FL

- Co-authored a multi-agent LLM system for longitudinal behavioral health coaching, accepted for publication at the IEEE/ACM CHASE 2026 conference (21% acceptance rate).
- Designed the system utilizing a vLLM inference engine and GPT-4o to decouple response generation from state management, reducing redundant conversational loops by 55%.
- Engineered a structured JSON-based state tracker and dynamic Python prompt-patching mechanism, achieving a 6.04/7.0 user experience score and consistent SMART goal adherence.

Software & Automation Intern

June 2025 – Aug. 2025

Merck & Co.

Rahway, NJ

- Designed and developed a Power Platform solution integrating Power Apps and Power Automate to streamline and support audit planning processes, improving team coordination and planning efficiency
- Automated team communications by leveraging the Microsoft Graph API to deliver real-time email and calendar reminders, directly enhancing deadline adherence.
- Ensured software reliability and maintainability by implementing automated test cases with Power Apps Test Studio and producing technical documentation for long-term scalability.

PROJECTS

Running Safer | runningsafer.com | *Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python, PostgreSQL*

- Built a full-stack web application featuring a responsive Next.js interface and Strava OAuth to securely authenticate users, import activity history, and render interactive route maps.
- Developed a FastAPI backend that decodes activity polylines, queries OpenStreetMap Overpass data, and processes pedestrian safety metrics like sidewalk availability to score individual street segments.

Thread Count | threadcountcarson.com | *Blazor Server, Blazorise, .NET 8 (C#), SQLite, Azure*

- Engineered a full-stack wardrobe management application using Blazor Server, implementing secure user authentication and persistent CRUD functionality via Blazor Identity and Entity Framework Core.
- Designed a modular UI with Blazorise components and automated cloud deployment to Azure App Service by establishing continuous integration and deployment (CI/CD) pipelines using GitHub Actions.

LEADERSHIP/INVOLVEMENT

Theta Tau Professional Engineering Fraternity | *Web Chair, FSU Liaison, Risk Chair*

Apr. 2024 – Present

Association for Computing Machinery | *Historian*

Aug. 2024 – Apr. 2025

TECHNICAL SKILLS

Languages: C++, Python, C#, JavaScript, TypeScript, SQL, Bash, HTML, CSS

Frameworks/Libraries: React, Next.js, .NET, Blazor, FastAPI, vLLM, Tailwind CSS, Vite

Developer Tools: Git, GitHub, Unix, Linux, Azure, Power Apps, Power Automate, Visual Studio Code